

Dynamic Single-Mode Lasers

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(Invited Paper)

Abstract—The 40-year history of research on semiconductor lasers for high-speed long-distance optical fiber communications, so-called dynamic single-mode (DSM) lasers, is reviewed. DSM lasers include phase-shift distributed feedback (DFB) lasers, uniform DFB lasers, wavelength tunable (WT) distributed Bragg reflector lasers, WT distributed reflector lasers, and external reflector lasers. The vertical cavity surface emitting laser is also a type of DSM laser that is applied for rather short-distance communication. Photonic-integrated circuits with monolithic integration of DSM lasers with other devices, as well as photonic crystal lasers, have advanced significantly to support stable operation of photonic subsystems. The DSM laser is currently essential for most long-haul optical communications and transoceanic submarine cables, as well as medium-distance local area networks. Its application to sensing systems is another interesting area.

Index Terms—Distributed Bragg reflector (DBR) laser, GaInAsP/InP laser, PIC, distributed feedback (DFB) laser, dynamic single-mode (DSM) laser, dynamic single-mode laser, external reflector laser, integrated laser, laser dynamics, optical fiber communication, photonic integrated circuit, ring resonator laser, vertical-cavity surface-emitting lasers (VCSEL), wavelength tunable (WT) laser, wavelength tunable laser.

I. INTRODUCTION

THE dynamic single-mode (DSM) laser is a type of semiconductor laser used in ultrahigh-speed and long-distance optical fiber communications. It has inherent properties such as 1) stable single-mode operation for superb spectral purity, 2) the wavelength tunability essential for communication technology, and 3) an operating wavelength in the low-loss transmission band of optical fibers. In addition, it exhibits the superior merits of semiconductor lasers in general, such as a long life, a small size, low power consumption, high efficiency, high-speed direct modulation, a wide wavelength coverage, low cost by mass production, and monolithic integration with other devices.

“Stable single-mode operation” is the most important property for overcoming the material and waveguide dispersion in single-mode fibers for high-speed transmission, increasing the capacity of wavelength division multiplexing (WDM), and enabling phase modulation schemes for wide-band communications. Stable single-mode operation should be robust against bias current variation and/or high-speed direct modulation, and temperature variation. “Wavelength tunability” is essential for wavelength control in high-density WDM and wavelength tun-

ing of local oscillator lasers in coherent systems. Finally, “An operating wavelength in a low-loss transmission band” is quite obviously necessary for longer-distance transmission, as well as for better power efficiency.

Before the details of research on DSM lasers are discussed, let us look back on the early days of semiconductor lasers and optical fiber communications. In 1958, Schawlow and Townes proposed “Optical maser” [1], and a few years later, the first lasers, namely ruby lasers [2], were demonstrated. Next, He–Ne gas lasers were demonstrated [3]. The first lasing operation of semiconductor “junction lasers,” i.e., GaAs lasers, was realized almost simultaneously and independently by Hall *et al.* [4], Quist *et al.* [5], and Nathan *et al.* [6]. Furthermore, GaAsP lasers were demonstrated by Holonyak, Jr., *et al.* [7] in 1962, after the semiconductor optical amplifier (SOA) was predicted by von Neumann in 1953 [8]. These junction lasers consisted of homo-junction structure. Therefore, lasing operation was achieved only under pulsed operation at room temperature (RT) owing to its high threshold current, except for continuous wave (CW) operation at low temperature [9].

Basic works on optical fiber communications covered a broad range of research topics, such as optical glass fibers with a cladding layer and high-speed external optical modulators [10]. Incidentally, light was used in communications in earlier days, for example, in the form of smoke signals, beacon signals, and, as a part of the communication network built during the era of Darius the Great in 500 BC, by the use of reflected sunlight. Visual signal transmission was used, for example, the semaphore system introduced by Chappe in 1792; furthermore, a telecommunication system with light carriers, known as the photophone, was demonstrated in 1880 by Bell. After the appearance of lasers, a He–Ne laser was used to demonstrate optical communications in the form of a “talking” light at Bell Telephone Laboratories in 1961 [11]. An optical “fiber” communication experiment, which used an optical glass fiber, was first demonstrated at the Tokyo Institute of Technology by the author and his students on May 26, 1963, as shown in Fig. 1 [12], [13].

The possibility of low-loss and wideband optical glass fiber communications was proposed by Kao and Hockham in 1966 [14]. Several fundamental properties of semiconductor lasers were studied, such as high-speed direct modulation [15] and operation under reproducible lasing mode by a narrow current stripe [16]; an understanding of mode competition was gained by developing dynamics theory [17], and the potential for optical integrated circuits was examined [18]. Incidentally, RT-CW operation of semiconductor lasers was first realized by Alferov *et al.* in 1969 [19] and then by Hayashi and Panish in 1970 [20]. To realize RT-CW operation, they used “hetero-junctions” instead of homojunctions with a GaAlAs/GaAs structure, which

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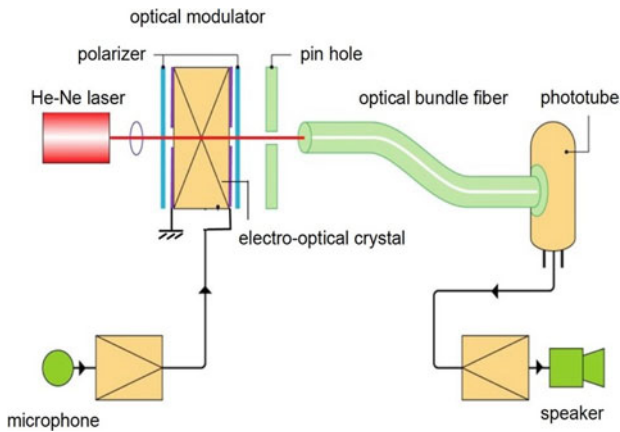


Fig. 1. Optical fiber communication was demonstrated in 1963 at Tokyo Institute of Technology [12], [13], where bundled optical glass fibers were used for light transmission, and a He–Ne laser, ADP crystal, and photo-tube were used as the light source, modulator, and detector, respectively.

had been proposed by Kroemer in 1963 [21]. In the same year, a fabrication process for low-loss silica fiber, called chemical vapor deposition, was developed by Kapron *et al.* at Corning [22], and these events also attracted industrial interest in optical fiber communications.

A semiconductor laser was thought to be attractive for optical fiber communications. However, the GaAlAs/GaAs lasers in the period just after the first RT-CW operation were unstable and far from suitable for efficient optical fiber communications, owing to their configuration as a current-confined gain-guiding waveguide with a Fabry–Perot (FP) resonator [23]. Thus, the concept of the DSM laser consisting of an active refractive index waveguide for stable transverse single-mode operation was suggested by the author and Yamada in 1972 [24], [25] as the first step. Next, the principle of DSM lasers, namely, longitudinal single-mode lasers consisting of two distributed reflectors connected by a phase shift of $\pi/2$, was proposed by the author and Hayashi in 1974 [26]. This structure solved the two mode operation problem theoretically predicted for refractive-index-coupled distributed feedback (DFB) lasers by Kogelnik and Shank in 1972 [27]. Before presenting their theory, they demonstrated a DFB dye laser in 1971 [28]. In the same year, Kaminow and Weber demonstrated distributed Bragg reflector (DBR) dye lasers [29]. Streifer *et al.* pointed out theoretically that single-mode operation in uniform DFB lasers could be attained by using additional axially asymmetric side facet mirrors with a proper phase shift, in 1975 [30]. GaAlAs/GaAs DFB lasers using optical pumping were demonstrated by Nakamura *et al.* in 1973 [31].

As the second step, integrated lasers were developed for monolithic integration of semiconductor lasers with a wavelength selection and/or tuning mechanism, as well as other optical components. Several GaAlAs/GaAs integrated lasers were demonstrated in 1975 and later [32]–[36]. Integrated twin-guide (ITG) lasers consisted of three-fold coupled resonators and realized single-mode pulsed operation [32]. As the third step, we had to realize laser diodes (LDs) operating in the 1.5- μm wavelength band to achieve the lowest loss transmission of silica

fibers in accordance with the development of low-loss fibers. The first RT-CW operation of 1.5- μm -band GaInAsP/InP LDs was achieved by Arai *et al.*, and then by Akiba *et al.*, Kawaguchi *et al.*, and Kaminow *et al.*, in 1979 [37]–[41]. Then, as the fourth step, the first GaAsInP/InP DSM-LD operating in the 1.5- μm band was demonstrated by Utaka *et al.* of the author's group in 1980–1981 [42]–[45]. The name “DSM laser” was chosen in 1981 [46]. For practical DSM-LDs, 1.5- μm -band uniform DFB-LDs with one side facet mirror were developed in 1981–1982 [47], [48] as DSM-LDs with thermal wavelength tuning by varying the laser mount temperature. The early development of DSM lasers is summarized in [49], and the potential of single-mode optical fiber communications at the lowest loss band is summarized in [50]. Phase-shift DFB-LDs, which became the de facto standard of DSM lasers, were demonstrated by Sekartedjo *et al.* in 1984 [51]. Phase-shift DFB-LD arrays are frequently used for wavelength tunable (WT) lasers in which thermal tuning is applied by changing the laser mount temperature.

As the fifth step, WT-LDs with sections for tuning both the reflector wavelengths (Bragg wavelengths) and the phase by refractive index variations were proposed by the author and Utaka in 1980 [52] as DSM-lasers. Their principle of tuning by injected carrier plasma was demonstrated by Tohmori and the author *et al.* in 1983 [53], and the theory was presented in 1984 [54]. WT-LDs with an extremely wide tuning range obtained using superstructure gratings (SSGs) and a sampled grating (SG) were developed by Tohmori and Yoshikuni *et al.* [55] and Jayaraman *et al.* of Coldren's group [56], respectively, in 1993. Coldren pointed out that the wide wavelength tunability was due to the Vernier effect. WT-LDs using thermal refractive index variations in both the resonators and phase sections [57], and external reflector LDs as an example [58], were developed for narrow spectral operations.

DSM lasers include phase-shift DFB-LDs, uniform DFB-LDs with one side facet mirror, and WT-DBR-LDs and WT distributed reflector (DR) LDs with Bragg wavelength and phase tuning sections, respectively. These DSM lasers are widely used in ultrahigh-speed and long-distance optical fiber communications. The vertical-cavity surface-emitting lasers (VCSELs) proposed by Iga [59]–[61], in which a phase shift between two DBRs occurs, are also DSM-LDs; they are applied in medium-distance optical communications because of their low power handling capability. Micro-cavity lasers, such as photonic crystal lasers [62], are another type of DSM-LDs and are expected to be used as the lasers in photonic integrated circuits (PICs). This research is discussed in Section II. Various early attempts that cannot be introduced here owing to space limitations can be found in [63]. Theoretical improvements and the development of material processes contributed greatly to improvements in the performance of DSM lasers, as discussed in Section III and IV, respectively; the applications are discussed in Sections V. The prosperity gained from the Internet may not have been possible without the development of optical fiber communications, in which DSM lasers act as a key device.

This paper reviews the 40-year history of research on DSM lasers.

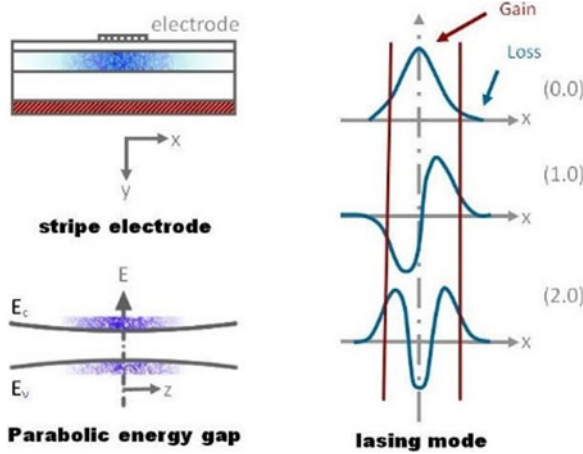


Fig. 2. Refractive index waveguide for transverse control of LDs, 1972 [24], [25].

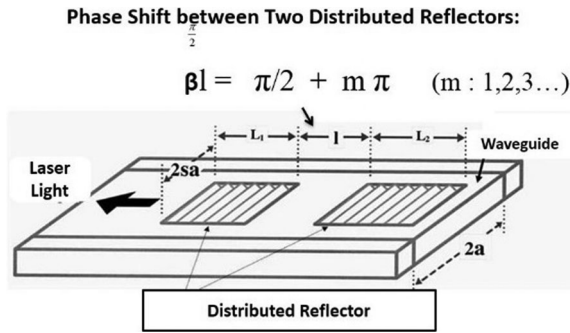


Fig. 3. Single-mode resonator with two distributed reflectors connected by phase shift of $\beta l = \pi/2$ or a quarter-wavelength, 1974 [26].

II. PROGRESS ON DSM LASERS

A. Concept and Principle of DSM Lasers

DSM-LDs for ultrahigh-speed and long-distance optical fiber communications must provide stable single-mode operation, a tunable wavelength, and a wavelength band for low loss transmission [64]. Early FP GaAlAs/GaAs LDs consisted of a current-confined gain-guide, such as that in [23], and sometimes fell into relaxation oscillation owing to strong nonlinearity at the resonance-like frequency near the upper limit of direct modulation [15]. Some of them also exhibited a kink in the output light owing to deformation of the transverse current distribution by the light intensity itself [23] from the stimulated radiation. To overcome the transverse mode instability, as the first step, in 1972 the author and Yamada proposed LDs in which the gain guide was replaced by a refractive index waveguide supporting favorably only the fundamental transverse mode, as shown in Fig. 2 [24], [25], which led to the suggestion of using a DSM laser for stable single-mode operation.

Then, in 1974, the author and Hayashi proposed a laser consisting of two distributed reflectors coupled by a phase shift with the goal of realizing single-longitudinal-mode operation, as shown in Fig. 3 [26].

At that time, composite resonators, short-cavity resonators, and distributed reflectors were considered as wavelength selec-

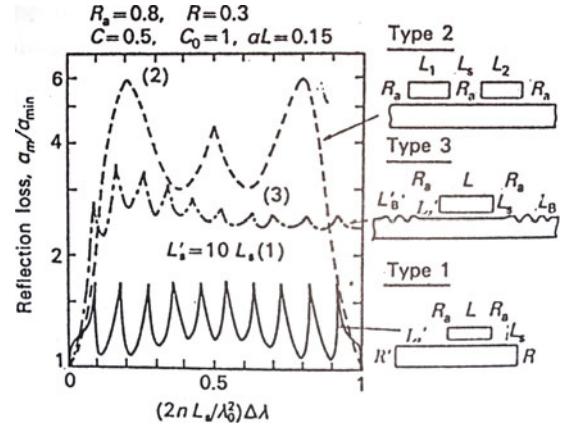


Fig. 4. Axial mode selectivity for several resonator configurations in integrated twin-guide (ITG) LDs, 1977 [71].

tive mechanisms. The composite resonator was well known to exhibit a sharper resonance by increasing the number of coupled resonators [32]; therefore, it was used in an attempt to realize single-mode operation of LDs (shown later in Fig. 4). Some of these composite resonators are currently used in DSM-LDs with, for example, lasers with external reflectors, as mentioned in a later section. The short-cavity laser was represented by VCSELs as low-power-consumption DBR lasers [59]. The distributed reflectors were commonly used later for DSM lasers such as DFB and DBR-LDs. Kogelnik and Shank demonstrated the first DFB dye lasers in 1971 [28], and they presented a noted theory of uniform DFB lasers [27], which yielded two-mode operation in refractive-index-coupled uniform DFB lasers despite operating in a single mode in the case of gain coupling. In the same year, Kaminow and Weber demonstrated DBR dye lasers [29]. Nakamura *et al.* reported the first optically pumped GaAlAs/GaAs DFB lasers in 1973 [31].

The author and Hayashi proposed theoretically, as mentioned above, that single-longitudinal-mode operation of LDs is available in lasers consisting of two refractive-index-coupled distributed reflectors connected to each other with a phase shift θ_{ps} of $\pi/2$ or $\lambda/4$ in length, in addition to integer multiples of π , as shown in Fig. 3 [26]:

$$\theta_{ps} = \pi/2 + \pi m \quad (m = 0, 1, 2, \dots). \quad (1)$$

Note that refractive index coupling is practical in conventional LDs. The condition of (1) for a laser consisting of two distributed reflectors with a phase shift of θ_{ps} is applicable either for active or passive guides. If both reflectors are active, the lasers are called phase-shift DFB-LDs [51]. If both reflectors are passive and the intermediate space is partly active and partly passive, the lasers are DBR-LDs. Finally, if one reflector is active and the other is passive, the lasers are called DR-LDs [89]. These all belong to the family of DSM lasers. The use of an asymmetric taper for the purpose of phase shifting was also proposed later by Haus and Shank [65] and also by Tada *et al.* [66].

A single-mode laser consisting of a uniform DR with axially asymmetric facet mirrors, here called a uniform DFB-LD, was proposed by Streifer *et al.* in 1975 [30]. In this case, the resonator

configuration is, in principle, one-half of the laser compared with that of two periodic reflectors, as shown in Fig. 3. Thus, the phase shift θ_{uf} between the facet mirror and the periodic reflector for single-mode operation is a half of θ_{ps} .

The wavelengths of these lasers without wavelength tuning sections are fixed, but they are tuned by changing the temperature of the laser mount to satisfy the requirement for wavelength tunability of DSM-LDs. In this sense, these lasers without tuning sections are also called wavelength-fixed DSM-LDs. For stable transverse-mode operation, a buried heterostructure (BH) was effectively used later [67], [79]. LDs with an external reflector, such as an external ring resonator, are also a type of DSM-LD [107], as are VCSELs [61] when the $\pi/2$ phase-shift that satisfies the requirement of (1) is put into the intermediate region between the two DBRs. Photonic crystal lasers also behave as DSM-LDs [62].

B. GaAlAs/GaAs DFB Lasers

Optically pumped GaAlAs/GaAs DFB lasers were demonstrated for the first time by Nakamura *et al.* in 1973 [31]. Uniform DFB-LDs based on GaAlAs/GaAs compounds emitting at 830 nm have been investigated extensively [68] and have influenced the development of DSM lasers. Single-mode operation under CW excitation as well as under relatively high-speed modulation were realized in 1976 [69] in uniform GaAlAs/GaAs DFB-LDs.

C. Demonstration of Integrated Lasers

Integrated lasers consisting of laser active waveguide sections monolithically combined with passive waveguide sections along which other optical components could be mounted, as shown later in Fig. 11, were demonstrated first with GaAlAs/GaAs compounds in 1975. These are ITG lasers (Suematsu *et al.* [32]), taper-coupled lasers (Merz *et al.* [33]), direct internal coupling lasers (Reinhart *et al.* [34]), and large optical cavity lasers (Hurwitz *et al.* [35]). Integrated lasers became a powerful tool for investigating new junction lasers, such as DR-LDs and WT-LDs, as well as PICs, as discussed in Sections II.G and II.J, respectively. So-called butt joint built-in (BJB) integrated lasers [36] are suitable for efficient coupling schemes using field distribution matching as well as equivalent refractive index matching between two waveguides with a high yield process; thus, they were frequently used for DSM lasers. Single-mode operation was demonstrated in ITG lasers with multiple resonances under pulsed operation in 1976 [70].

Axial mode selectivity for various resonator configurations using ITG LDs is illustrated theoretically in Fig. 4 [71]. The mode selectivity for multiple resonances of FP resonators is comparable to that of distributed reflectors. The multiple resonator configuration shows comb-like wavelength selectivity and is used in lasers with external resonators, as discussed in Section II-G.

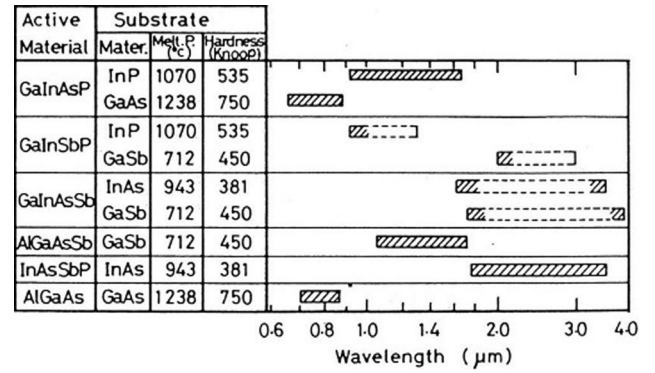


Fig. 5. Prospective material systems for long-wavelength regions, in 1976 [50].

D. Room-Temperature CW Operation of 1.5-μm-Band GaInAsP/InP Long-Wavelength Lasers

In the mid-1970 s, Keck *et al.* predicted that the lowest loss band in silica optical fibers lies near the long-wavelength region around 1.4 to 1.7 μm [72], and the material dispersion could be zero at 1.3 μm [73]. After these works, laser materials for the long-wavelength band attracted attention. There were two possibilities for crystal systems expected to match the prospective lowest loss wavelength region of silica fibers, GaInAsP/InP and AlGaAsSb/GaSb, as shown in Fig. 5. The GaInAsP/InP crystal system was selected because of its higher melting temperature and the greater hardness of its substrate (InP) compared with GaSb [50]. Hsieh *et al.* demonstrated 1.1 μm GaInAsP/InP lasers in 1976 [74]. Then, 1.3 μm GaInAsP/InP LDs were demonstrated by Oe and Sugiyama [75] and Yamamoto *et al.* [76]. Use of (100) InP substrate, which is commonly used today, was proposed by Itaya *et al.* in 1977 [77].

In 1979, Arai *et al.* of the author's group achieved RT-CW operation of 1.5-μm-band GaInAsP/InP LDs after overcoming the melt-back problem in liquid phase epitaxy [37], [38], followed by Akiba *et al.* [39], Kawaguchi *et al.* [40], and Kaminow *et al.* [41] in the same year. The lowest loss wavelength band of silica fiber, with a loss of 0.2 dB/km, was realized in the 1.55 μm wavelength band [78] in the same year.

Transverse fundamental mode operation was realized in a BH structure [67], which was originally developed for GaAlAs/GaAs LDs but exhibited an Al oxidation problem; it became fully successful in, for example, GaInAsP/InP DSM-LDs without Al content [79], [80].

E. Demonstration of DSM Lasers in 1.5-μm Band

The author was extensively involved in the development of DSM lasers at the lowest loss wavelength band of silica fibers in the 1.5-μm band. A GaInAsP/InP long-wavelength DBR-LD was developed in 1978 [81], [82]. DSM laser operation at 1.3 μm was confirmed in 1980 by Sakakibara *et al.* [42]. The first DSM-LDs, i.e., DBR-ITG-LDs, emitting at 1.5 μm were realized by Utaka *et al.* of the author's group in 1981 [43], [44] (Fig. 6); next, CW operation [45] and DSM operation at up to 3 GHz [83], which was close to the relaxation frequency, were

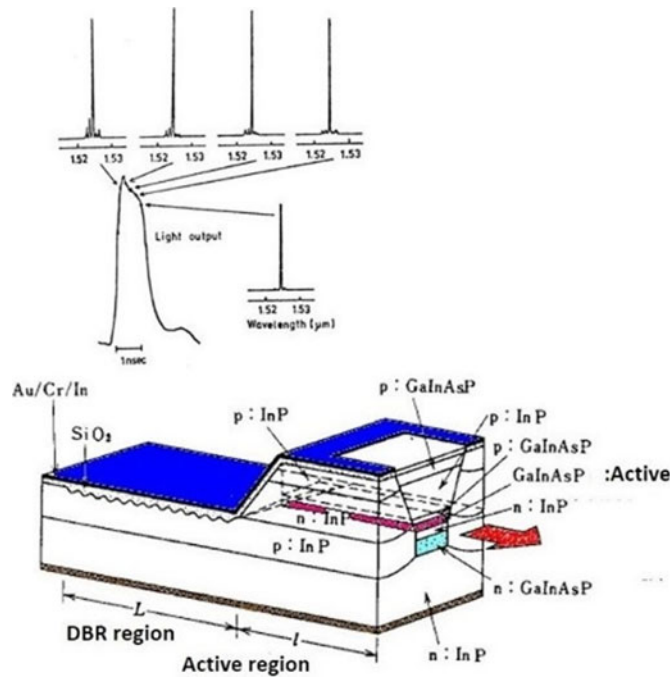


Fig. 6. DSM operation of 1.5- μm GaInAsP/InP LDs, 1981 [43].

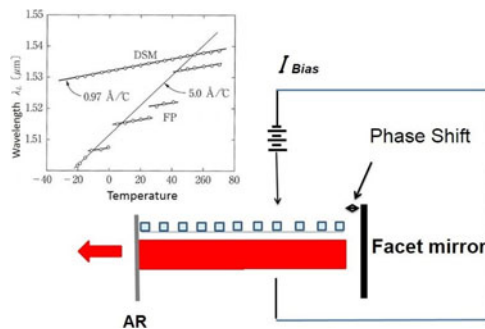


Fig. 7. 1.5- μm -band uniform DFB-LDs with one side facet mirror on back side, acting as a wavelength-fixed DSM laser, 1981 [47]. Anti-reflection coating was applied to front facet.

realized. The author had insisted on testing DSM operation with rapid modulation up to the relaxation frequency, where nonlinearity often degraded the single-mode characteristic.

After the demonstration of the DSM-LD in the 1.5 μm band in 1981, it attracted much attention. Further, 1.5- μm -band DSM-LDs were developed extensively as a light source for ultra-high-speed and long-distance optical fiber communications at the lowest loss wavelength band, as summarized in [49]. The term “DSM laser” was first used in [46].

F. DSM Lasers With Thermal Tuning by Changing Laser Mount Temperature: DFB-LDs

Utaka *et al.* developed 1.5- μm -band uniform DFB-LDs with one side facet mirror for commercial use as a DSM laser with thermal wavelength tuning in 1981 (see Fig. 7) and demonstrated DSM operation at a modulation speed of 500 Mbits/s [47]. Matsuoka *et al.* also reported DSM operation of 1.51- μm -band uniform BH-DFB-LDs in 1982 [48]. After these successful re-

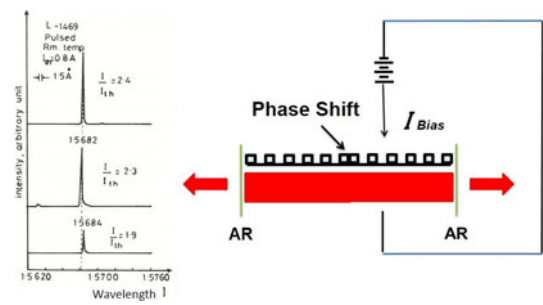


Fig. 8. 1.5- μm -band phase-shift DFB-LDs acting as wavelength-fixed DSM-LDs, 1984 [51]. Anti-reflection (AR) coatings were applied to both facets.

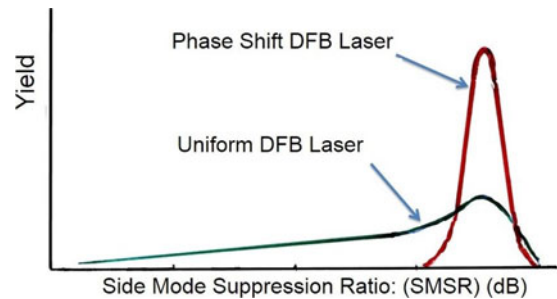


Fig. 9. Rough sketch on comparison of production yields for phase-shift DFB-LDs and uniform DFB-LDs. Courtesy of Mitsubishi Electric Ltd.

sults, wideband optical fiber transmission experiments using 1.5–1.6- μm -band DSM lasers have spread worldwide [84]–[86].

Incidentally, immediately after the first demonstration of 1.5 μm GaInAsP/InP DSM-LDs [43], Koyama *et al.* found a “dynamic wavelength shift” in 1981 [83], which was strongly related to the transmission bandwidth of optical fibers, as discussed in Section III.

Sekartedjo *et al.* of the author’s group realized 1.5- μm -band GaInAsP/InP phase-shift DFB-LDs, which acted as high-yield DSM-LDs with thermally tuned wavelengths, in 1984 (see Fig. 8) [51]. Phase-shift regions at the center of cavity, as shown in Fig. 8, were realized by electron beam direct writing [51], and later by combining negative and positive photoresists [87].

Uniform DFB-LDs with one side facet mirror (see Fig. 7) were considered for higher efficiency despite their lower production yield owing to the uncertainty in the facet mirror position relative to the reflector periodicity [88]. In contrast, phase shift DFB-LDs (see Fig. 8) are used frequently because of their high production yield despite their lower efficiencies, as light is emitted from both facets. The latter are also used frequently for laser arrays because of their high yield potential. The production yields for uniform DFB-LDs and phase-shift DFB-LDs are roughly compared in terms of the side-mode suppression ratio (SMSR) in Fig. 9. The SMSR is defined as the power of the lasing mode S_0 relative to that of the suppressed side mode S_1 , S_0/S_1 , to specify DSM-LDs, as discussed in Section III. The SMSR of a typical DSM-LD is around 40 dB. High-performance phase-shift DFB/DBR LDs, which consist of an active periodic reflector and a passive periodic reflector and are called DR-LDs, were also developed [89], [90].

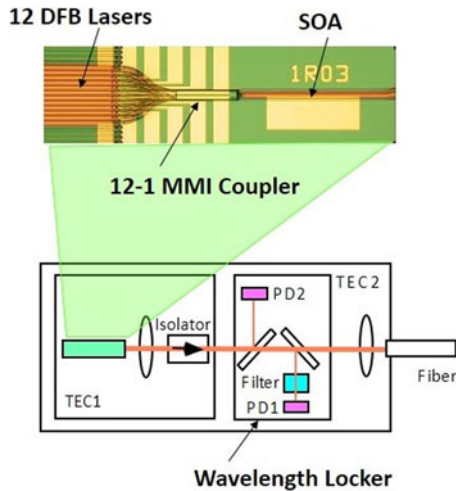


Fig. 10. Temperature tunable laser array consisting of 12 phase-shift DFB-LDs, 2001 [94], the DFB laser array was proposed in 1995 [92]. Courtesy of Nippon Telegraph and Telephone Co. (NTT) and NTT Electronics Co. Ltd. (NEL).

The wavelengths of phase-shift DFB-LDs and uniform DFB-LDs are fixed under a constant heat sink temperature, but they are effectively tuned by changing the temperature of the laser mount, as mentioned previously [91]–[94]. The multi-wavelength DFB laser array was proposed in 1995 [92]. Phase-shift DFB-LD arrays with temperature tuning are monolithically integrated with a multimode interferometer combiner and SOA to produce a wavelength-selectable module. External modulations occur for high-speed applications [93]. Because wavelength selection for a channel spacing of 100 or 50 GHz is required, the lasing wavelength of phase-shift DFB-LDs has been controlled by a thermoelectric cooler and an etalon filter (wavelength locker), as shown in Fig. 10 [94]. An integrated module covering 97 channels with a 50-GHz spacing (a total wavelength range of 38 nm) was demonstrated for WDM applications. A laser chip greater than 1 mm of cavity length yields a narrow spectral width of a few hundred KHz. Phase-shift DFB-LD arrays are also sometimes called WT-DFB-LDs [95].

G. Wavelength Tunable LDs as DSM Lasers: WT-DBR-LDs

During the attempted demonstration of DSM-LDs in the $1.5 \mu\text{m}$ band, the author and Utaka proposed WT-DBR-LDs with separate reflector wavelength (Bragg wavelength) tuning sections and a phase tuning section with refractive index variations obtained by electrical means, as shown in Fig. 11, in 1980 [52]. WT-LDs with injected carrier plasmas were first demonstrated in 1983 by Tohmori and the author *et al.* with a BJB-DBR-LD monolithically integrated into the laser together with a phase tuning section, for simplicity, as shown in Fig. 12; a continuous wavelength tuning range of 0.4 nm was achieved [53]. The theory of WT-DBR-LDs with reflector wavelength tuning sections and a phase tuning section was presented in 1984 [54]. Here, electric currents are injected into each tuning section of the WT-DBR-LD so that the refractive index of each section is varied by the plasma effect of injected electrons. Thus, the

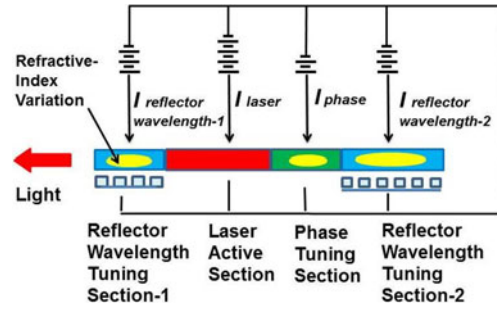


Fig. 11. Principle of WT-DBR-LDs with separate reflector wavelength (Bragg wavelength) tuning sections and phase tuning section with refractive index variations obtained by electrical means, 1980 [52]; by injected carrier plasma, 1983 [53] 1984 [54]; and by temperature variation, 1992 [57].

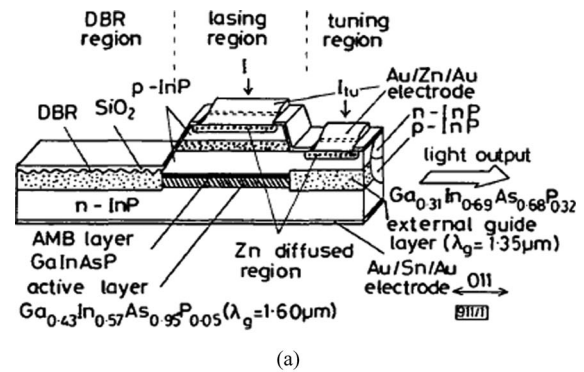


Fig. 12. First demonstrated WT-LDs with electronic phase tuning yielding continuous wavelength tuning of 0.4 nm, 1983 [53]. The word of “AMB” stands for “anti-melt-back”.

lasing wavelength is tuned continuously by the currents of the tuning sections. During the wavelength tuning process, the reflector wavelength tuning and phase tuning should be interrelated to satisfy Eq. (1). Constant-wavelength operation within a temperature range of more than 40°C was also demonstrated. Subsequently, WT-DBR-LDs with electrodes for both Bragg reflector sections and a phase tuning section were demonstrated by

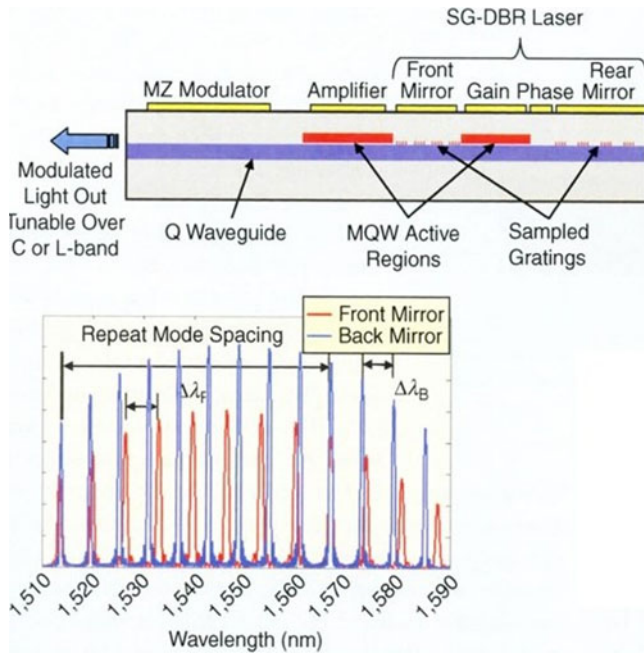


Fig. 13. Widely WT DBR-LD with two SGs having several different resonant wavelength pitches, 1993 [56]. Courtesy of Prof. L. A. Coldren.

Murata *et al.* in 1987 [96]. Such multi-segmented WT-DBR-LDs were extensively developed [97]–[100].

In 1993, WT-DBR-LDs based on SSG-DBR-LDs, which had a much wider tuning range, were demonstrated by Tohmori, Yoshikuni, *et al.* [55], and SG-DBR-LDs were developed by Jayaraman *et al.* of Coldren's group, as shown in Fig. 13 [56], with wide wavelength tuning ranges of over 100 nm obtained by carrier tuning. Various WT-LDs, such as twin-guide structures [97], Y-junction waveguides for different grating pitches [98], and multicontrol-electrode WT-LDs [99]–[101], were developed.

The tuning wavelength ranges of WT-DBR-LDs consisting of simple DBRs are limited by the tuning rates of the refractive index variations in the tuning sections. However, the tuning widths of WT-LDs with SSG-DBRs or SG-DBRs are wider by one order of magnitude, owing to the so-called Vernier effect, as Coldren pointed out. He commercialized WT-LDs with wide tuning range [102], [103]. The high reflectance wavelengths of the SSG-DBR or SG-DBR are multiple and comb-like, as shown in Fig. 13 [56], in contrast to the single wavelength peak for a simple DBR. The pitches of the high reflectance wavelength at both reflectors are set to differ slightly, as shown in Fig. 13. Thus, the lasing mode is made to hop from one mode to another by slight current tuning of one side reflector, resulting in a wider wavelength tuning range.

WT-LDs with separate Bragg wavelength tuning sections and a phase tuning section, and refractive index variations caused by temperature changes due to electrical heaters, were also developed by Woodward *et al.* [57], [104], [105], such as that in Fig. 14 [106]. The wavelength tuning mechanisms here are essentially identical to that of tuning by plasma effects (see Fig. 11). Those WT-LDs with thermo-tuning are specified to

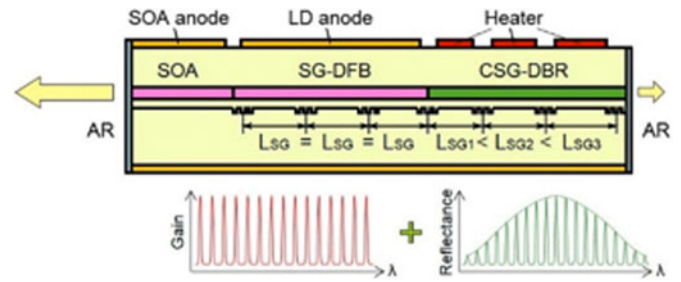


Fig. 14. WT chirped SG DR (WT-CSG-DR) LDs in which the local temperatures is changed by three local heaters in the DBR region, 2010 [106]. Courtesy of Sumitomo Electric Industries, Ltd.

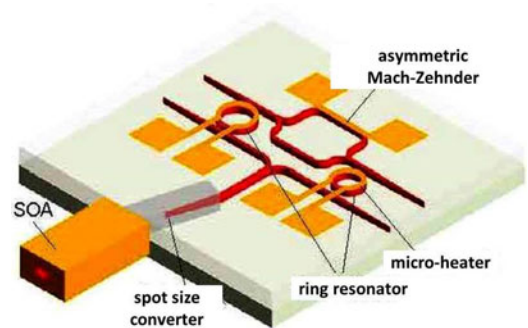


Fig. 15. Narrow spectral WT LD with external ring resonator, in 2013 [108]. The original words in Japanese on above figure are rewritten into English by the author.

have a narrow spectral width to meet the requirements of phase modulation systems.

Ring resonator lasers were developed as widely tunable WT-LDs, e.g., [107], as well as for narrow spectral width (see Fig. 15) [108]. WT-LDs with external reflectors or filters consisting of liquid crystal mirrors are specified to have a narrow spectral width because they have a longer cavity and also lack the shot noise of electron injection [58].

External reflector lasers with an extremely wide tuning range were realized [109]. Lower power consumption and a higher production yield overall, including the temperature-controlled power, are specific to WT-LDs.

H. VCSELs as DSM-LDs

VCSELs were proposed by Iga in 1977 [59]; RT-CW operation was realized by Koyama *et al.* of Iga's group in 1988 [60] and later by another group [110]. They are distinctive for their compactness (see Fig. 16) and two-dimensional integration. The early development of the VCSEL is summarized in [61]. The VCSEL is also a DSM-LD when the phase shift described by (1) is installed between the two DBRs in the proper operating wavelength band. WT-VCSELs (Chang-Hasnain [111]) and athermal wavelength VCSELs (Janto *et al.* [112]) were extensively developed for high-performance operation [113]. VCSELs are studied intensively for medium- and short-distance optical communication such as data links and Ethernet. A single-mode VCSEL is a powerful tool for sensor applications.

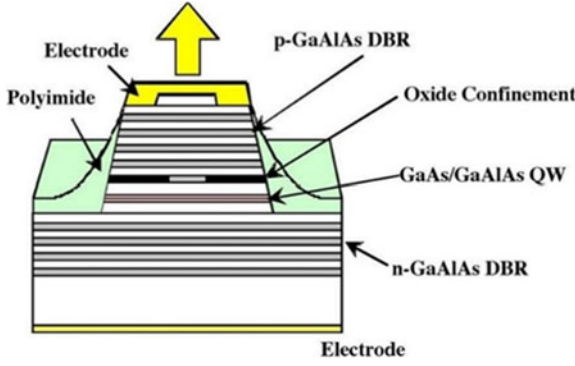


Fig. 16. VCSEL: LD emitting light perpendicular to the substrate with small power consumption, 1977 [59], 1998 [60].

I. DSM Lasers with Superlattice Structures

In addition, progress in the basic performance of DSM-LDs was also accelerated significantly in 1980–1990 by the use of quantum wells (QWs) as active layers. The first trials of conventional QW lasers were made by van der Ziel *et al.* in 1975 [114] using optical pumping, and then by Rezek *et al.* in 1978 using current injection [115]. In 1983, Osbourn *et al.* proposed a strained QW to improve the optical gain in QWs [116], and Adams proposed band structure engineering for low-threshold-junction lasers in 1986 [117]. The QW technology yielded a lower threshold, higher power, better temperature characteristics, and higher speed than a bulk active layer [118]. Most recent DSM lasers use QWs as active layers. To further enhance the laser performance, quantum wires and quantum dots (QDs) were proposed by Arakawa and Sakaki [119] in 1982 and were expected to yield a higher gain [120]. Self-assembled QDs exhibited good performance [121]. The first laser operation with QDs was achieved by Hirayama *et al.* in 1994 [122]. A wire-like laser was also demonstrated [123]. Temperature-insensitive operation of QD-LDs was favorably demonstrated in 2004 [124]. Numerous works were performed using superstructures to increase the transmission bandwidth of modulated lasers, which are discussed in Section III.

J. Photonic Integrated Circuits: Advanced DSM Lasers

PICs, [125], consisting of monolithically integrated optical active devices [126], such as waveguide modulators, SOAs, and lasers [32]–[36] have been extensively developed for optical communications with fairly large-scale integration [127]–[130]. Many practical DSM lasers have been monolithically integrated with a modulator, SOA, and other components [92], [93], [94], [131]. The waveguide optical isolator [132], which is a key component for high-performance PICs and silicon photonics [133], is under development. Photonic crystal [134], [135] has been extensively developed due to its potentiality of compact monolithic integration. RT-CW operation of photonic crystal lasers with extremely low threshold currents was recently demonstrated; they are expected to be used in future compact integrated circuits [62], [136].

III. DYNAMICS OF DSM-LDS

A. Theory of Dynamics and Side-Mode Suppression Ratio (SMSR)

A theoretical understanding of semiconductor lasers is essential for obtaining their ultimate performance. A theory of junction lasers for spontaneous and stimulated emission was presented in 1964 by Lasher and Stern, who introduced the band tail model [137]. A dynamic theory of junction lasers was given by Nishimura *et al.* in 1970 [138], [139], who introduced carrier relaxation with density matrix analysis. This theory was later extended by Yamada and the author [140]. However, this analysis was limited to above the lasing threshold. Therefore, it became essential to connect the performances below and above the threshold [141]. A spontaneous emission factor represents part of the intensity from spontaneous emission into a specific lasing mode, which was theoretically given by the author and Furuya in 1977 [142],

$$C_s = \xi \frac{\lambda^4}{2\pi^2 n_{eq}^3 V_a \Delta\lambda_m} \quad (2)$$

where, ξ is the energy confinement factor for the waveguide in the cavity, n_{eq} is the effective refractive index of the cavity, λ is the resonant wavelength, V_a is the volume of the cavity, and $\Delta\lambda_m$ is the spectral width of the spontaneous emission specific to the lasing material.

Asada and the author derived rate equations including the linear and third-order nonlinear gain [143]. The excess gain suppression for the non-lasing mode is very small compared with the gain of the lasing mode, by $\tau_c/\tau_s \approx 10^{-4}$, where τ_c is the carrier relaxation time. And τ_s is the lifetime of the spontaneous emission. Therefore, stable single-mode operation of semiconductor lasers owes much to the mode selectivity of the resonant cavity structure.

The SMSR is given by Koyama *et al.* [144] (see Fig. 17) as the ratio of the light intensity of the lasing main mode S_0 divided by that of suppressed side mode S_1 . The SMSR for static operation is given as [145]

$$\text{SMSR} = \frac{S_0}{S_1} = \frac{\eta_d}{\xi C_s} \frac{\Delta\alpha_{\text{loss},1}}{\alpha_{\text{loss},0}} \frac{I - I_{\text{th}}}{I_{\text{th}}} \quad (3)$$

with

$$\Delta\alpha_{\text{loss},1}/\alpha_{\text{loss},0} = (\alpha_{\text{loss},1} - \alpha_{\text{loss},0})/\alpha_{\text{loss},0} \quad (4)$$

where η_d is the differential quantum efficiency, ξ is the energy confinement factor for the active waveguide, $\Delta\alpha_{\text{loss},1}/\alpha_{\text{loss},0}$ is the relative loss difference of the cavity between the suppressed side mode and the lasing main mode, C_s is the spontaneous emission factor given by Eq. (2), and I and I_{th} are the injection and threshold currents, respectively. The SMSR deteriorates by several dB when it is modulated rapidly up to the modulation limit of the relaxation frequency [15]. An early numerical analysis was made [146]. As can be seen in Fig. 17, a mirror loss difference $\Delta\alpha_{\text{loss},1}$ larger than a few tens of cm^{-1} is required to achieve an SMSR larger than 40 dB; such a laser is regarded as a DSM laser. The term SMSR to specify single-mode operation [Eq. (3)] was filed as a Japan Industrial Standard (JIS

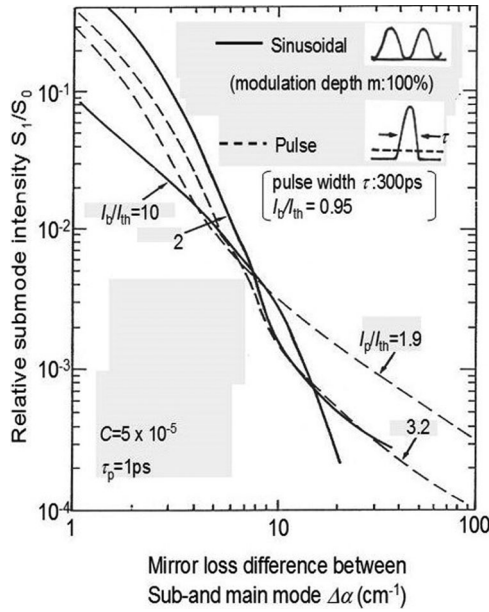


Fig. 17. Inverse of SMSR (side mode suppression ratio S_0/S_1) versus mirror loss difference between sub (side) mode and main mode, 1984 [144], [145].

0594041) in 1989 and internationally with the IEC (IEC60747–5–1; 6.2.7.8) in 2002, representing the wide use of the concept of DSM lasers.

B. Direct Modulation and Reduction of Resonance-like Phenomena

Ikegami and the author [15] identified the upper limit of the direct modulation frequency f_r , which is called the resonance-like frequency or relaxation frequency. This resonance-like phenomenon is closely related to the self-pulsation at microwave frequencies of early semiconductor lasers [147]. A reduction in the ultimate modulation speed even under high-bias conditions was also discussed in relation to the nonlinear gain [148]. This sharp resonance or nonlinearity at f_r is fairly well suppressed by setting the active guide width nearly equal to the carrier diffusion length; a few μm [149]. Nonlinear distortion in the analog modulation was described in [150]–[152]. An enhanced maximum modulation bandwidth was presented in [153], [154]. Owing to this large nonlinearity at the resonance-like frequency, single-mode operation in DSM lasers was strictly tested at near the upper modulation frequency limit.

C. Dynamic Spectral Width and Transmission Band Width of Optical Fibers

The development of DSM lasers enabled the investigation of the detailed spectral behavior of semiconductor lasers. Koyama *et al.* reported spectral broadening under high-speed direct modulation in 1981 [83]. This phenomenon causes a dynamic linewidth $\Delta\lambda_s$ owing to the modulation of the carrier density in the active region. This dynamic wavelength shift becomes larger when the modulation speed is faster or the modulation depth is higher, as confirmed by a large signal modulation analysis [155].

Shortly after this, the dynamic spectral width was given by the time derivative of the photon density S for a simple expression of the wavelength chirp [156], [157] with the linewidth enhancement factor $\alpha = \Delta n'/\Delta n''$ [158], where $\Delta n'$ and $\Delta n''$ are the real and imaginary parts of the refractive index variation ($\Delta n = \Delta n' - j\Delta n''$), respectively:

$$\Delta\lambda_s = -(\lambda^2/4\pi c)(1/S)(dS/dt)\alpha. \quad (5)$$

The transmission bandwidth B at the bit rates of a single-mode optical fiber of length L is given by the α value of the laser or external modulator and the dispersion of a single-mode fiber $d^2\beta/d\omega^2$, as follows [157].

$$B = \left(2\sqrt{2\sqrt{\alpha^2 + 1} + 2\alpha\sqrt{(d^2\beta/d\omega^2)L}}\right)^{-1} \quad \alpha \geq -1/\sqrt{3}. \quad (6)$$

Assuming that $\alpha = 4$ and the dispersion is 20 ps/nm/km, $B\sqrt{L}$ becomes 25 Gbps·km^{1/2}. Numerous studies were conducted to reduce the value of α , and the introduction of superlattice structures was found to be effective for this purpose [159]–[162], especially in a quantum box (or QD), which was predicted to make the value of α zero at the gain peak wavelength [159]. In fact, frequency chirping in external modulators also was found to be expressed by α , and the advantages of electro-absorption type and Mach–Zehnder-type modulators were pointed out because α can be reduced to 0 to -1 [163]. Hence, the possible transmission bandwidth of a standard single-mode fiber can be increased by a factor larger than 2 by using these external modulators. A proposal was made to reduce the value of α using strained QWs [164], [165]. Other approaches to reducing the chirping were made by introducing special cavity structures combining a DFB and a passive DBR [166], [167].

As discussed in Sections III-B and C, the transmission bandwidth of optical fibers is described by the linewidth enhancement factor for both the direct modulation case and external modulation case. The external modulation scheme is advantageous for high capacity-long distance transmission in comparison with the direct modulation scheme because of the smaller linewidth enhancement factor in the external modulation [168]. Monolithic integration of an external modulator with a DSM laser as described in Section II-J is an attractive solution to expand the bandwidth of the optical fiber transmission. A variety of DSM lasers with a monolithically integrated modulator has been commercialized.

D. Intrinsic Spectral Width of Laser Diodes

The spectral width of junction lasers is important for coherent communications and was measured by Okoshi and Kikuchi in 1980 [169]. The spectral linewidth of a semiconductor laser under CW operation $\Delta f_{sw,sta}$, was given theoretically by Fleming and Mooradian in 1981 [170] and then by Henry with the α parameter [158]. This linewidth is rewritten in a simplified form using the spontaneous emission factor C_s given in (2),

$$\Delta f_{sw,sta} = \frac{C_s N_{th}}{4\pi\tau_s S'} (1 + \alpha^2) = \frac{\xi \lambda^4 \ln(1/r_f r_r)}{16\pi^3 n_{eq}^2 \Delta\lambda_m L_{cav} V_a} \frac{1 + \alpha^2}{I/I_{th} - 1} \quad (7)$$

where α is the linewidth enhancement factor defined previous section, S' is the photon number in the cavity, N_{th} is the carrier number in the cavity at the threshold, τ_s is the carrier lifetime, r_f and r_r are the front and rear reflectivity of the cavity reflectors, respectively, L_{cav} is the length of the cavity, I is the injection current, and I_{th} is the threshold current. The typical spectral width of single-mode semiconductor lasers is given by Eq. (7) as several tens kHz to a few MHz, depending especially on the cavity length. The spectral width tends to be narrower by longer cavity as shown by Eq. (7), including the laser with external resonator, such as ring resonator [108]. Considerably narrower spectral width was achieved by electrical feedback [171].

In WT-LDs with thermal tuning, appreciable additional spectral width broadening is not found, as mentioned previously [105], [172]; however, in WT-LDs with injected plasma tuning, the additional spectral widths Δf_t are at a somewhat appreciable level, as given by Amann and Schimpe [173]:

$$\Delta f_t = 4\pi e (c^2/\lambda^4) \left\{ (\partial\lambda/\partial I_p)^2 I_p + \kappa L_B (\partial\lambda/\partial I_B)^2 I_B \right\} \quad (8)$$

where κ and L_B are the coupling coefficient and the length of the DBR, respectively, and $\partial\lambda/\partial I_p$ and $\partial\lambda/\partial I_B$ are the tuning efficiency of the phase and DBR sections, respectively. The additional spectral width is calculated, for example, as a few to several MHz.

Extra noise due to reflected waves was also discussed in relation to the external reflection point (e.g., [174]); therefore, an optical isolator is essential for WT-LDs with a narrow spectral width. The theoretical noise figure of the SOA dominates in QDs and approaches the theoretical limit of 3 dB [175].

IV. MATERIALS FOR DSM LASERS

The properties of the semiconductor materials used in DSM-LDs are important for their effect on the lasing wavelength, lasing performance, device structure, and fabrication processes. High-quality epitaxial growth techniques for semiconductor materials are also indispensable.

A. Standard Materials

We can use InP substrates with an epitaxially grown GaInAsP active layer for wavelengths of 1300 nm to 1600 nm. This material system is developed for application to silica-fiber optical communications, as discussed in Section II [37]–[41], and is widely used for practical lightwave systems. Nonradiative recombination, especially for 1.5- μ m-band GaInAsP, is specified to reduce the total recombination efficiency by increasing the temperature [176]–[180]. AlGaInAs was also developed for these wavelength ranges and is advantageous for carrier confinement, which improves the temperature characteristics of lasers.

GaN is a wide gap material in blue-UV light sources [181], [182]; however, it is interesting for the formation of superlattices

in deep wells [114]–[120], which could be applicable to the infrared transition.

B. Strained Quantum Wells and Quantum Dots

In very thin layers such as QWs, we can use lattice-mismatched materials without serious crystal deterioration. Recent lasers use strained QWs to improve their performance. GaInAs is used to extend the wavelength range of GaAs. The dilute nitride system of GaInNAs is attractive for further elongating the wavelength [183]. Because the band discontinuity between GaAs and GaInNAs is large, a combination of these materials is interesting for achieving long-wavelength lasers and good temperature characteristics. Wavelength widening up to 1300–1600 nm on GaAs is becoming possible.

The small three-dimensional structure of dots allows an increase in the lattice mismatch. An important dot fabrication technique is self-organized formation caused by the Stranski–Krastanov mode [120]. InGaAs dots on GaAs have been developed for emission wavelengths in the 1000–1500 nm range [184], [185].

C. Epitaxy Techniques

Early lasers were developed by liquid phase epitaxy. The controllability and applicability of various materials and compositions with fine heterostructures have been realized by organic-metal vapor phase epitaxy (OMVPE) [186] and molecular beam epitaxy. The quality of GaInAsP crystals obtained by OMVPE was improved by applying low-pressure conditions [187]. OMVPE is the main production technique for DSM lasers.

V. DSM LASER APPLICATION IN PHOTONICS SYSTEMS

A. Optical Fiber Communications

The amount of digital data traffic is rapidly increasing as the Internet grows. DSM lasers in the 1.5 μ m band, represented by phase-shift DFB-LDs, uniform DFB-LDs, WT-DBR-LDs, WT-DR-LDs, and external reflector LDs, are used for the most part in long-haul transmission worldwide. DSM-LDs have come into practical use, for example, in the late 1980s for overland transmission by uniform DFB lasers, in the early 1990s for trans-Pacific submarine cables using phase shift DFB lasers, and in the mid-2000 s for dense wavelength division multiplexing (DWDM) systems using WT-LDs (see Fig. 18). Significant advances in the system configuration have been made, such as modulation schemes, routers, optical fiber amplifiers, equalizers, modulators, switches, optical circuits, optical isolators, and high-speed electronic devices.

DSM-LDs were also applied for the sending end of “fiber to the home (FTTH)” in around 2001. External modulation schemes have been used for most long-distance systems, whereas direct modulation schemes are used for rather short-distance systems. Those are mostly intensity modulation systems, the so-called first-generation systems. Medium-distance (~ 10 km) local area networks in Ethernet format are used for

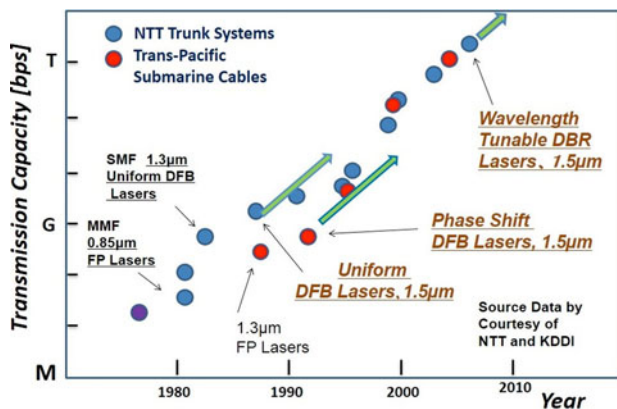


Fig. 18. DSM-LDs in long-distance optical fiber communication. Annual increase in transmission capacity per fiber is shown, courtesy of source data from Nippon Telegraph and Telephone Co. (NTT) and KDDI Corporation. The words of "SMF" and "MMF" stand for "single-" and "multi-" mode fibers, respectively.

intra-campus networks or building backbones, where VCSELs are also used.

One problem with DWDM systems is the large number of signal channels. WT-LDs provide a solution and ensure network performance. The yearly adoption of WT-LDs for the network use has been approximately 10 times increased in five years from 2005. As a result, WT-LDs are used in around 60% of 10 Gbps transponders [188]. More advanced signal formats for coherent optical transmissions [189]–[191], called second-generation systems, are already in use [192], [193]. These systems require many optical components, such as WT-LDs, as well as optical modulators, balanced detectors, and lasers acting as reference oscillators. Therefore, in future optical fiber communication systems, it is very important to realize an integrated optical transmitter that includes the above functions in one chip.

B. Other DSM Laser Applications

DSM-LDs are currently used in various applications from industrial to personal applications. Stable linewidth lasers are used to sense many parameters such as gas, position, and distance. By extending this sensing function, optical coherent tomography is gaining attention in the medical field. As the characteristics of lasers improve and new functions are invented, the application areas of DSM-LDs will increase in the future.

VI. CONCLUSION

Research on DSM-LDs, including phase-shift DFB-LDs, uniform DFB-LDs, WT-DBR-LDs, WT-DR-LDs, external reflector LDs, and VCSELs, is reviewed, in terms of their role in high-performance optical communications and information technology. The true nature of DSM-LDs was evident fairly early in their development and might ensure their use for optical fiber communications. Current optical communications are undergoing a new turn from first-generation systems to second-generation systems, in which DSM-LDs act as a core technology. We still expect the appearance of new system concepts, optical devices, and electronic devices. As a light source, the DSM-LD must provide more flexible wavelength control with

low power consumption and narrow spectral linewidth using photonic integrated circuits.

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